

## Integrantes

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## Laboratório 2 - Redes de Comunicação I

### 1 – PCs e VLANs do Laboratório

Utilizamos as VLANs de acordo com as instruções do laboratório, o modelo final, após a alteração do PC3, ficou da seguinte maneira:

- PC1 – VLAN 10 (VLAN A)
- PC2 – VLAN 20 (VLAN B)
- PC3 – VLAN 20 (VLAN B)
- PC4 – VLAN 20 (VLAN B)
- PC5 – VLAN 10 (VLAN A)
- PC6 – VLAN 20 (VLAN B)

### 2 – Saídas de show vlan brief (Antes e Depois da mudança do PC3)

*SW1 - Antes da mudança*

```
Switch>show vlan brief
```

| VLAN Name               | Status | Ports                                                                                                                                                                       |
|-------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 default               | active | Fa0/5, Fa0/6, Fa0/7, Fa0/8<br>Fa0/9, Fa0/10, Fa0/11, Fa0/12<br>Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Gig0/1<br>Gig0/2 |
| 10 VLAN_A               | active | Fa0/1, Fa0/3                                                                                                                                                                |
| 20 VLAN_B               | active | Fa0/2, Fa0/4                                                                                                                                                                |
| 1002 fddi-default       | active |                                                                                                                                                                             |
| 1003 token-ring-default | active |                                                                                                                                                                             |
| 1004 fddinet-default    | active |                                                                                                                                                                             |
| 1005 trnet-default      | active |                                                                                                                                                                             |

```
Switch>
```

## SW2 - Antes e Depois, SW2 não muda as portas de VLAN

Switch>show vlan brief

| VLAN Name               | Status | Ports                                                                                                                                                                                     |
|-------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 default               | active | Fa0/3, Fa0/4, Fa0/5, Fa0/6<br>Fa0/7, Fa0/8, Fa0/9, Fa0/10<br>Fa0/11, Fa0/12, Fa0/13, Fa0/14<br>Fa0/15, Fa0/16, Fa0/17, Fa0/18<br>Fa0/19, Fa0/20, Fa0/21, Fa0/22<br>Fa0/23, Gig0/1, Gig0/2 |
| 10 VLAN_A               | active | Fa0/1                                                                                                                                                                                     |
| 20 VLAN_B               | active | Fa0/2                                                                                                                                                                                     |
| 1002 fddi-default       | active |                                                                                                                                                                                           |
| 1003 token-ring-default | active |                                                                                                                                                                                           |
| 1004 fddinet-default    | active |                                                                                                                                                                                           |
| 1005 trnet-default      | active |                                                                                                                                                                                           |

## SW1 - Depois da mudança

Switch#show vlan brief

| VLAN Name               | Status | Ports                                                                                                                                                                       |
|-------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 default               | active | Fa0/5, Fa0/6, Fa0/7, Fa0/8<br>Fa0/9, Fa0/10, Fa0/11, Fa0/12<br>Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Gig0/1<br>Gig0/2 |
| 10 VLAN_A               | active | Fa0/1                                                                                                                                                                       |
| 20 VLAN_B               | active | Fa0/2, Fa0/3, Fa0/4                                                                                                                                                         |
| 1002 fddi-default       | active |                                                                                                                                                                             |
| 1003 token-ring-default | active |                                                                                                                                                                             |
| 1004 fddinet-default    | active |                                                                                                                                                                             |
| 1005 trnet-default      | active |                                                                                                                                                                             |

## 3 – Saídas de show interfaces fa0/24 switchport

Interface fa0/24 SW1 e SW2, respectivamente.

```
Switch# show interfaces fa0/24 switchport
Name: Fa0/24
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

```
Switch# show interfaces fa0/24 switchport
Name: Fa0/24
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
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Access Mode VLAN: 1 (default)
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Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```



## 4 – Resultados de Ping dentro da mesma VLAN e entre VLANs diferentes

Ping do PC2 192.168.10.12 (VLAN B) -> PC6 192.168.10.16 (VLAN B) ✓

Ping do PC2 192.168.10.12 (VLAN B) -> PC1 192.168.10.11 (VLAN A) ✗

```
C:\>ping 192.168.10.16

Pinging 192.168.10.16 with 32 bytes of data:

Reply from 192.168.10.16: bytes=32 time<1ms TTL=128
Reply from 192.168.10.16: bytes=32 time=1ms TTL=128
Reply from 192.168.10.16: bytes=32 time=1ms TTL=128
Reply from 192.168.10.16: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Request timed out.
```

## 5 – Explicação das Perguntas

O conceito de VLAN permite separar uma rede física em várias redes lógicas independentes. Dispositivos na mesma VLAN, mesmo conectados a switches diferentes, conseguem se comunicar porque pertencem à mesma rede lógica.

Já dispositivos em VLANs diferentes, mesmo em LAN física iguais, estão em redes lógicas diferentes, portanto não se comunicam diretamente, sendo necessário um roteador para realizar o roteamento entre elas.

## 6 – Eventuais Problemas Resolvidos

Encontramos um problema na **Parte 2**, pois a instrução diz que deve configurar as VLANs do Switch 2, para o PC4, PC5 e PC6.

Mas com a topologia utilizada, o PC4 não está conectado ao SW2, e sim no SW1. Então a configuração necessária do PC4, deverá ser realizada no CLI do SW1.